

Abelian

$ab = ba \rightarrow \text{group} \rightarrow \mathbb{Z}$

$\frac{\langle ab=ba \rangle}{\mathbb{Z} \times \mathbb{Z}}$

$S^1 \vee S^1$

$T = \mathbb{Z} \times \mathbb{Z}$

$T \hookrightarrow \mathbb{Z} \times \mathbb{Z}$

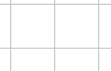
$K(\cdot, \mathbb{Z} \rtimes \mathbb{Z})$

[illegible]

R/P: 0-cell: 1
1-cell: 1
2-cell: 1

$ab \mid b = 0$

$a \mid b \mid (ab = b^n)$


 $\hookleftarrow$
 $\mathbb{R} \times \mathbb{R}^2 \times \mathbb{R}^2$
 $\cup \mathbb{R} \times \mathbb{R}^2 \times \mathbb{R}^2$
 $\cup (0, 0, \mathbb{R})$
 $(\frac{1}{2}, \frac{1}{2})$
 $-1 \leq z \leq 1$


Hand-drawn sketches of various organisms on grid paper:

- A segmented worm-like creature with five rounded segments.
- A starfish with five arms inside an oval boundary.
- A nautilus-like shell with concentric spiral lines.
- A mollusk-like creature with a large, rounded body and a small head.
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